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OpenShift Virtualization

Storage architecture & OpenShift Data Foundation

Red Hat

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Agenda

What we cover

- How KubeVirt consumes Kubernetes storage
- PVCs, access modes, and live migration
- ODF stack: block, file, object at a glance
- Sizing, performance, snapshots, operations
- Design checklist and documentation links

Learning objectives

Takeaways

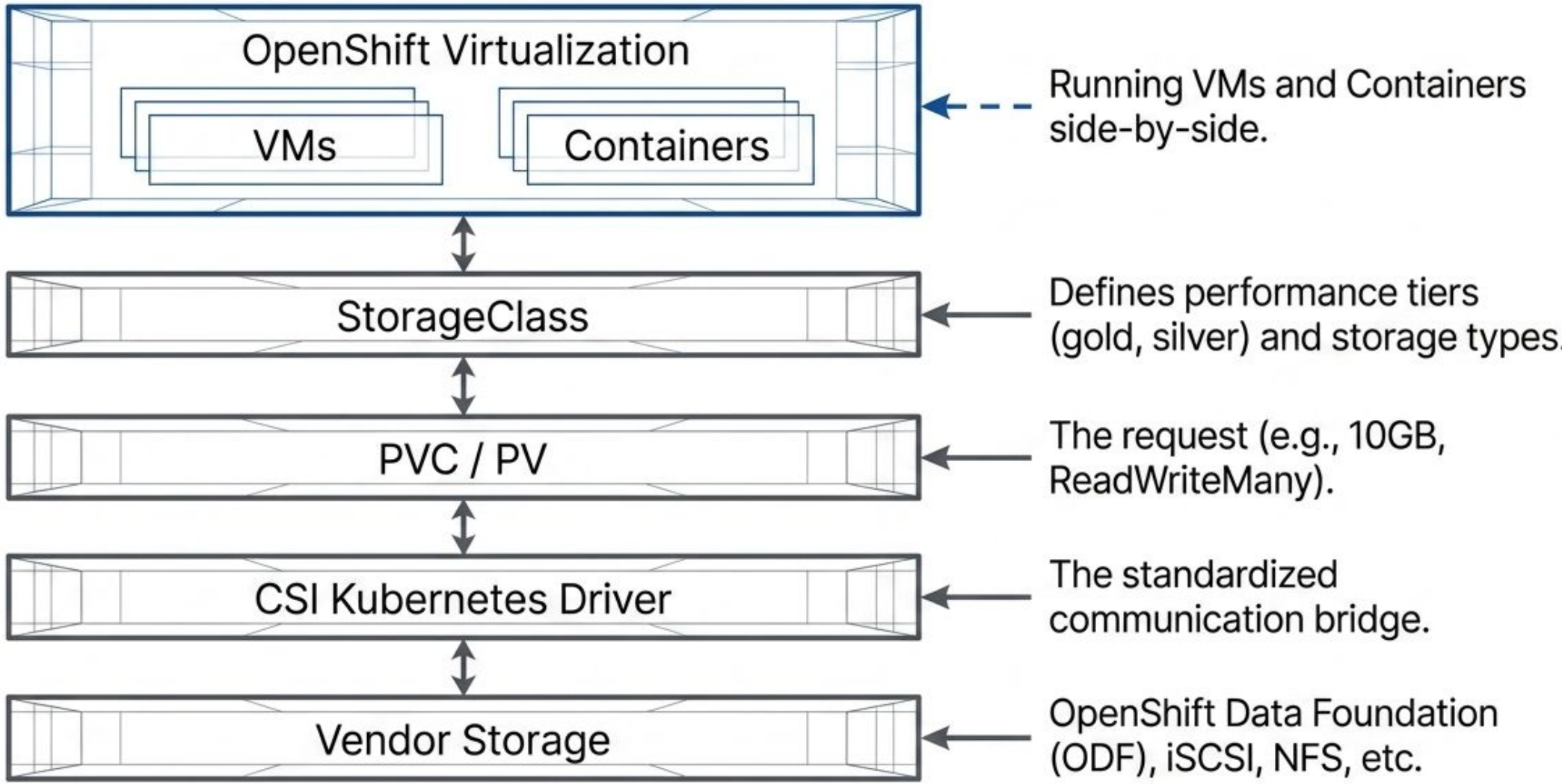
- Map VM disks to PVCs and CSI drivers
- Choose RWO vs RWX for boot and data disks
- Explain why migration needs shared storage classes
- Position ODF services for virtualization workloads

KubeVirt storage model

The Virt-to-Cloud Rosetta Stone

Legacy Concept	Kubernetes Translation
vSphere Datastore →	Persistent Volume (PV) & Persistent Volume Claim (PVC) **Note:** A PV represents physical storage. A PVC attaches it to a VM. **ReadWriteMany** (RWX) allows simultaneous mounting.
Dynamic Resource Scheduling (DRS) →	Pod Eviction Policies & Descheduler **Note:** Actively balances resources by live-migrating VMs to appropriate nodes based on utilization.

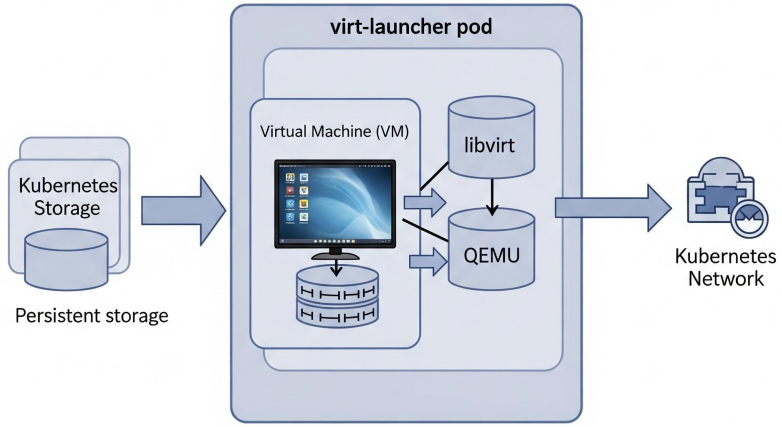
The CSI Abstraction Layer



Volumes and the virt-launcher pod

Same primitives as stateful apps

- Data volumes reference PVCs in the VM spec
- CSI provisions backing LUN or file share
- virt-launcher mounts volumes for QEMU
- Ephemeral disks use emptyDir or container images



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Access modes matter

RWO, RWX, and migration

- RWO: single node writer, typical block volumes
- RWX: many readers/writers, usually file or NFS
- Live migration needs shared access to disk data
- Match mode to StorageClass and Ceph pool type

The Golden Rules of ODF Virtualization

Rule 1: Live Migration Requires RWX

```
accessModes: [ "ReadWriteMany" ]
```

Note: PVCs must be mounted by multiple OpenShift nodes simultaneously to seamlessly hand off the VM state during a live migration.

Rule 2: Block Mode is Mandatory for Performance

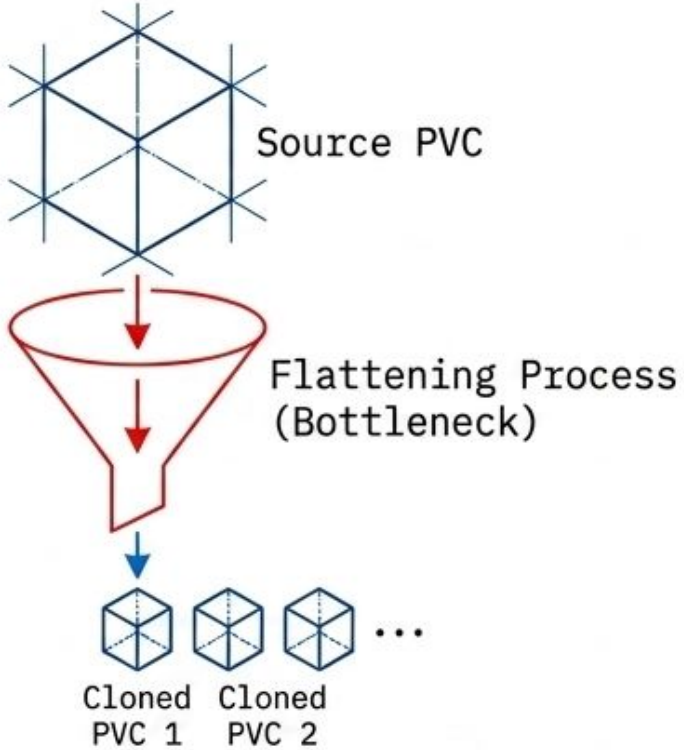
```
volumeMode: Block
storageClassName: ocs-storagecluster-ceph-rbd
```

Note: Ceph RBD (Block) volumes are vastly preferable to CephFS for raw VM disk performance.

The Cloning Strategy Matrix: Scaling Volumes

Method A: csi-clone (The Default)

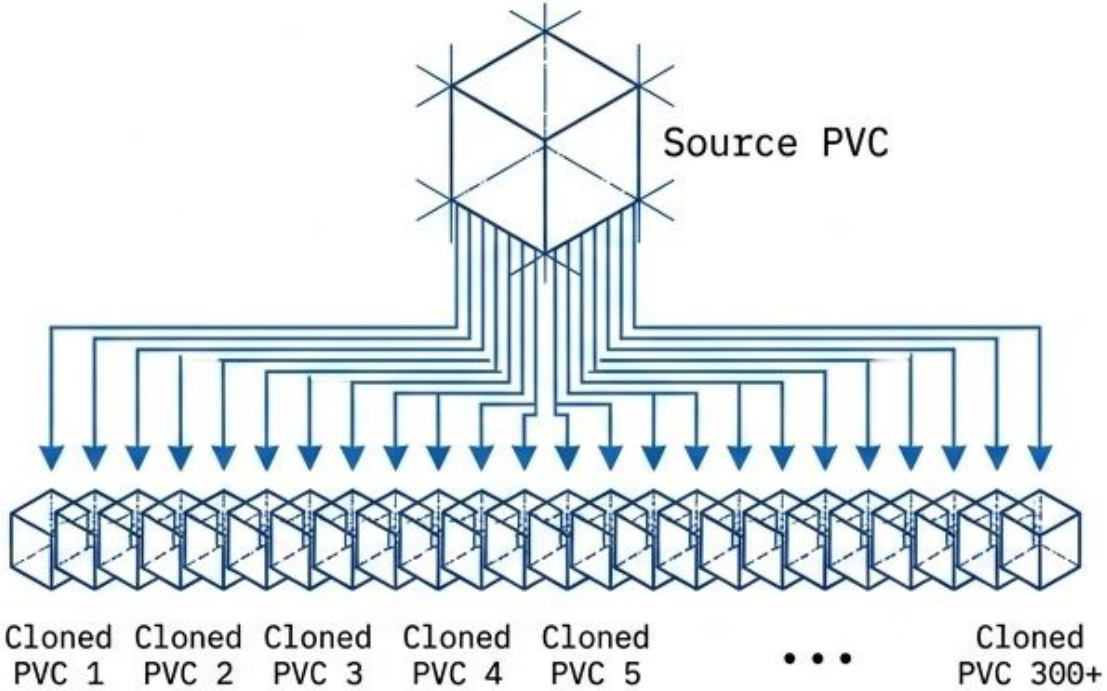
Mechanism: Standard storage profile default.



The Bottleneck: Initiates a background flattening process. When cloning hundreds of VMs from a single source PVC, performance degrades severely.

Method B: VolumeSnapshot (The Solution)

Mechanism: Uses ODF's CSI VolumeSnapshotClass.



The Advantage: Bypasses flattening delays. Recommended for mass-provisioning hundreds of clones from a single source instantly.

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Disk sources in brief

Common patterns for VM images

- PVC: persistent OS and data disks
- Container disk: image in container rootfs
- Cloud-init / config drive PVCs
- Hotplug volumes for data disks
- Container disk not for production DB roots
- PVC backed by fast block for latency
- Use golden-image pipelines to PVCs
- LUN passthrough for special cases only

Live migration & HA

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Live migration requirements

Shared visibility of the same volume

- Source and target nodes must reach the same PVC
- Storage must allow attach on the new node
- RWX file shares simplify concurrent access patterns
- Test migration in each StorageClass you deploy

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Performance and placement

Latency, IOPS, and node topology

- Keep storage network close to compute nodes
- Size pools for peak IOPS, not average only
- Spread OSDs across failure domains
- Watch virt-launcher I/O wait during load tests

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ODF logical architecture

Operators, Ceph, CSI

- ODF operators deploy and manage Ceph cluster
- RBD block volumes via openshift-storage.rbd.csi
- CephFS for RWX file workloads
- NooBaa / MCG for object and buckets

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Names vary slightly by ODF version

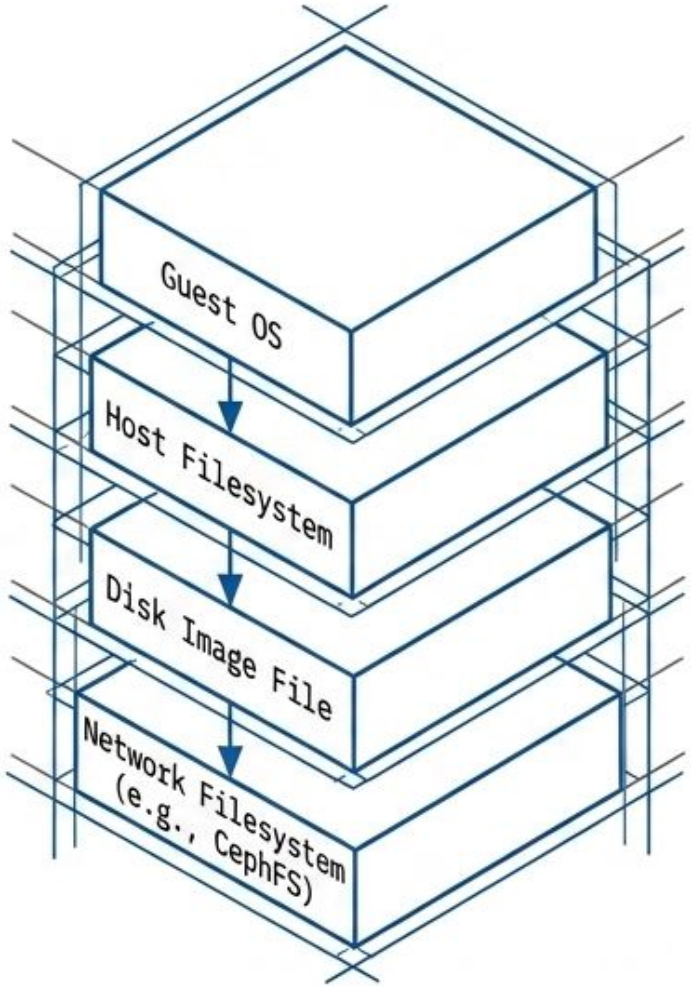
Block vs file for VMs

Typical mapping

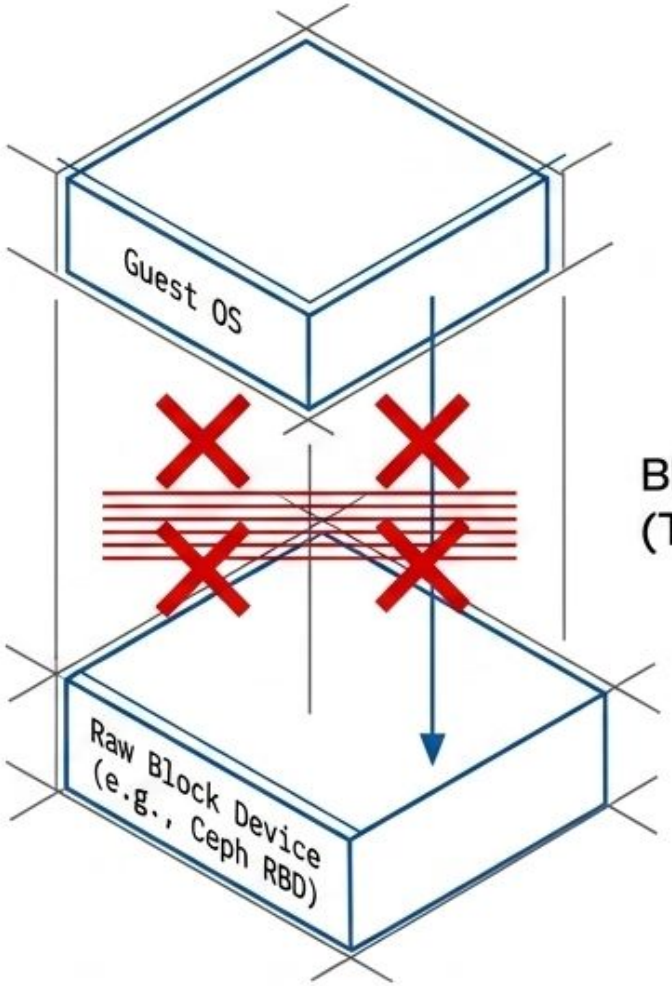
- Ceph RBD: low-latency VM root disks
- Thin provisioning and snapshots
- Clone golden images to new PVCs
- Encryption at rest with cluster keys
- CephFS or NFS RWX: shared data disks
- Some app clusters need shared file
- Higher metadata overhead than RBD
- Tune MDS and standby ranks for scale

The Performance Matrix: Filesystem vs. Block

Filesystem Stack
(The Bottleneck)



Analysis: Unnecessary storage layers degrade IOPS and introduce latency.



Block Stack
(The Standard)

Analysis: Bypasses the host filesystem layer completely. Directly mounts the RBD volume.

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StorageClasses and quotas

What users bind to

- StorageClass points to CSI driver and pool
- Default class simplifies self-service PVCs
- ResourceQuota and LimitRange per namespace
- Monitor ceph capacity and PG health

Snapshots, clones, and backups

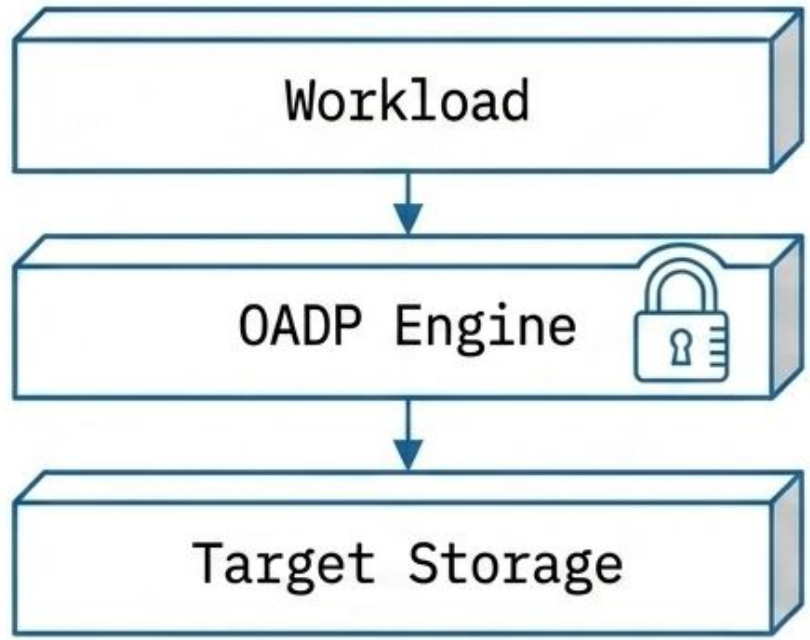
Data services for VMs

- VolumeSnapshot API for crash-consistent copies
- Clone PVCs for fast golden-image rollout
- Pair with OADP or vendor backup for DR
- Test restore of full VM definitions

RPO/RTO are business decisions

The Backup Paradigm: OpenShift API for Data Protection (OADP)

The Mandate: OADP safeguards customer applications at the namespace level—protecting Kubernetes objects, internal images, and VM persistent volumes.



Key Capabilities:

Self-Service (OADP 1.5+)

Namespace admins can execute backups without cluster-admin roles.

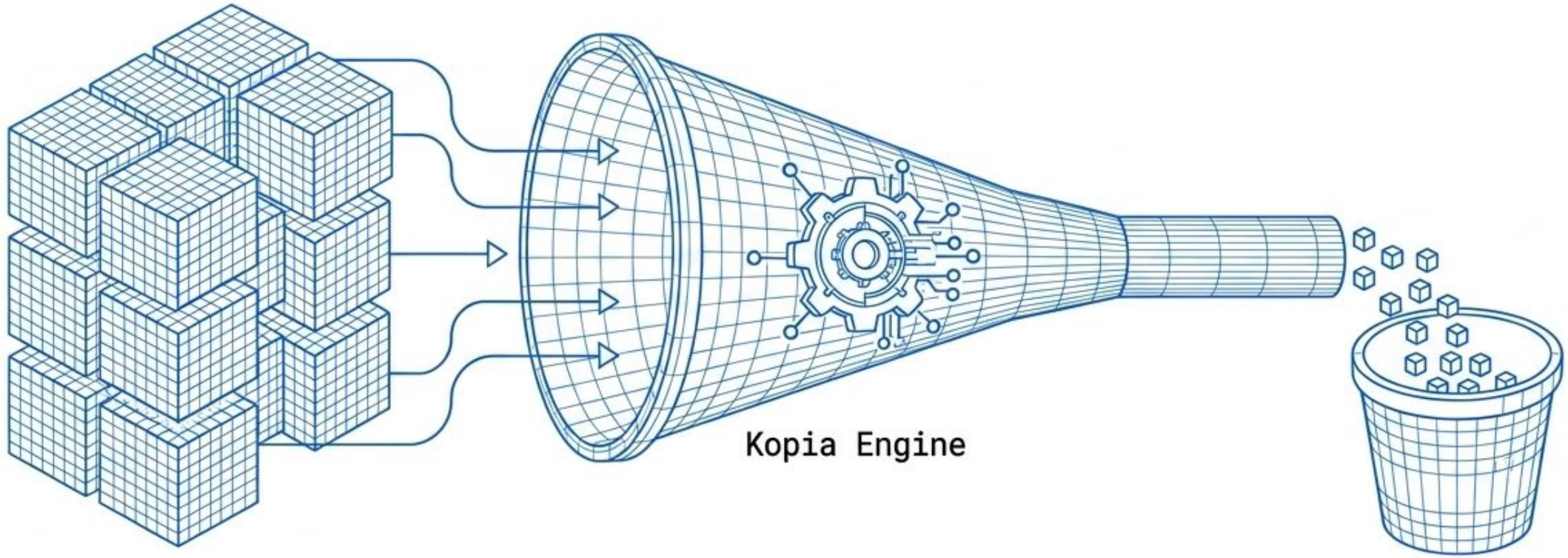
Target Storage

Backs up directly to S3-compatible object stores via Object Bucket Claims (OBCs).

Supported Targets

Multicloud Object Gateway (MCG), Ceph RADOS Gateway, AWS S3, MinIO.

The Incremental Backup Funnel (OADP Data Mover)



Step 1: CSI Snapshot

Leverages the Kubernetes CSI snapshot API to instantly capture the Block or Filesystem PV state.

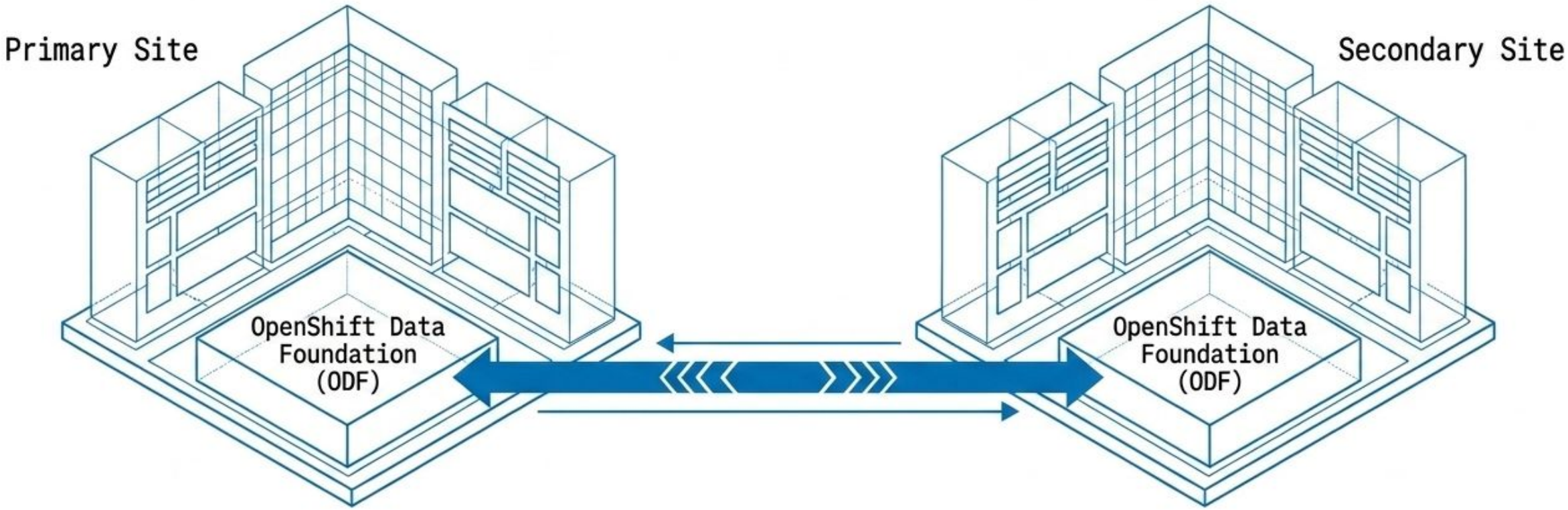
Step 2: Kopia Deduplication (The Uploader)

The OADP Data Mover reads the snapshot data and passes it through Kopia.

Step 3: Unified Object Repository

Writes highly optimized, deduplicated, and incremental backup payloads into the MultiCloud Object Gateway (MCG).

Cluster-Wide Protection: Metro-DR Architecture



Mechanism

Two-way synchronous data replication between managed OpenShift Virtualization clusters.

Components

Integrates Red Hat Advanced Cluster Management (RHACM) and OpenShift Data Foundation.

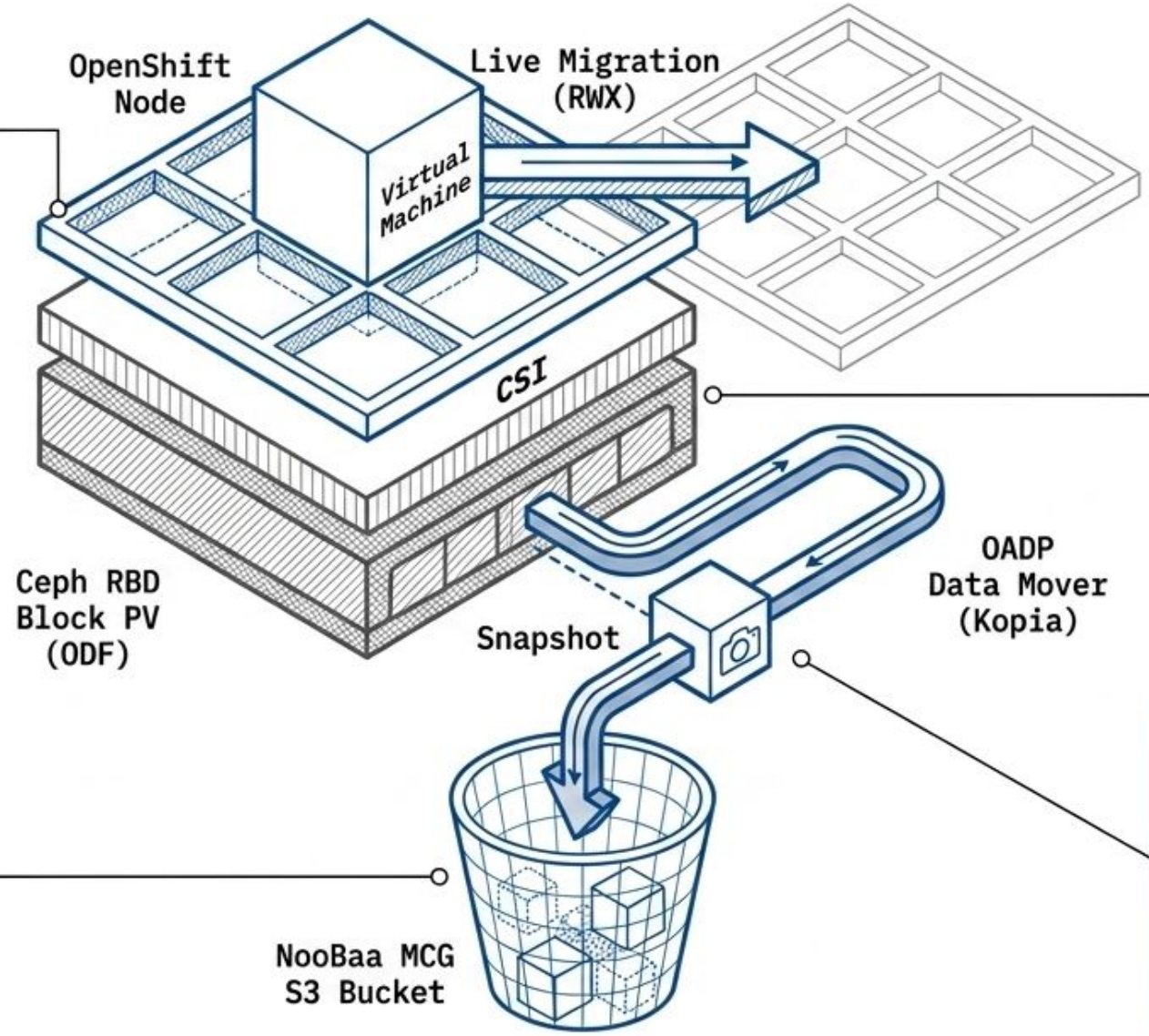
The Result

Ensures application failover from the primary to the secondary site with zero data loss, enabling rapid relocation post-disaster.

Synthesis: The Unified Data Fortress

Compute

OpenShift Node running a Virtual Machine payload.



Storage - ODF

Backed by a Ceph RBD block PV (provisioned via CSI), utilizing RWX for live migration readiness.

Protection - OADP

An orchestrated CSI snapshot processed by Kopia, seamlessly exporting incremental data to an MCG-backed object store.

The Insight

These are not disparate tools; they are tightly integrated, code-defined layers of a single data survival strategy.

Design checklist

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Pre-production checklist

Validate assumptions early

- Document StorageClass per workload tier
- Run migration tests on every disk class
- Size Ceph for rebuild and failure domains
- Plan network MTU for storage replication

Include networking and app teams

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Troubleshooting hints

When VMs fail to start or migrate

- oc describe pvc and pod for attach errors
- Check VolumeAttachment objects on nodes
- Review ceph health and slow requests
- Confirm SELinux and multipath on bare metal

Thank you

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